

Modeling and comparison of the Halbach array with different segments per pole

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Abstract. In this paper the Halbach array with different segments per pole are modeled and compared. The unified harmonic model for the magnetic flux density distribution of the Halbach array with different segments per pole is derived by using Fourier series and magnetic scalar potential. The coil power dissipation and higher harmonics are compared with the Halbach arrays, respectively. The dimension parameter of magnet array is optimized on the basis of minimum coil power dissipation by taking the magnetically levitated planar motor with moving magnets for example. The genetic algorithm is used to minimize the higher harmonics in z -component of the magnetic flux density distribution. Then the dimension parameters of magnet arrays are obtained, according to that the arrays are compared on the ratio between the higher harmonics and analytical model of z -component of the magnetic flux density distribution and the peak value of z - and x -component of the magnetic flux density distribution. The Halbach array with 3 segments per pole is chosen on the basis of the above comparison.

Keywords: Halbach array, harmonic model, genetic algorithm, power dissipation, planar motor

1. Introduction

The Halbach array is extensively used in various industries. It has the following advantages: The fundamental field is stronger than with a conventional array. The array does not require back-iron. The back-side magnetic field of the array is very low.

The principle of Halbach arrays was first proposed by Halbach [1]. Then Trumper et al. [2] recognized the use of this on linear motor and presented several geometries of such array in Cartesian, polar, and cylindrical coordinates. Due to the strong flux density, Jang et al. [3] applied a linear Halbach magnetized mover to an eddy-current braking system for high speed. Hoburg [4] introduced linear Halbach permanent magnet arrays for both levitation and propulsion in a low-speed magnetic levitation (maglev) vehicle. By comparison with conventional magnet array, Jian et al. [5] used Halbach magnetized permanent magnets in coaxial magnetic gears and obtained good performance. Jiang et al. [6] presented a planar motor which is constructed by four linear Halbach array segments. Not only the linear Halbach array, but also the plane Halbach array is used. Compter [7] described an electro-dynamic planar motor with moving coils and Halbach permanent array. Jansen et al. [8] replaced the linear Halbach array with the plane one to become a new linear motor. Then, they [9] proposed planar motors with moving magnets shown in Fig. 1, in which the Halbach magnets array is applied. Jin et al. [10] presented a novel linear and rotary permanent magnet motor by using the plane Halbach array.

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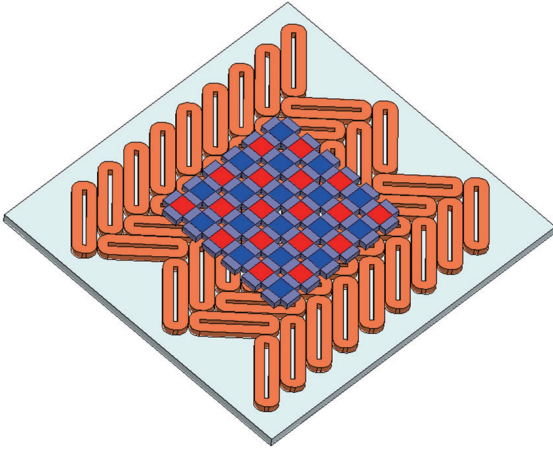


Fig. 1. Magnetically levitated planar motor with moving magnet.

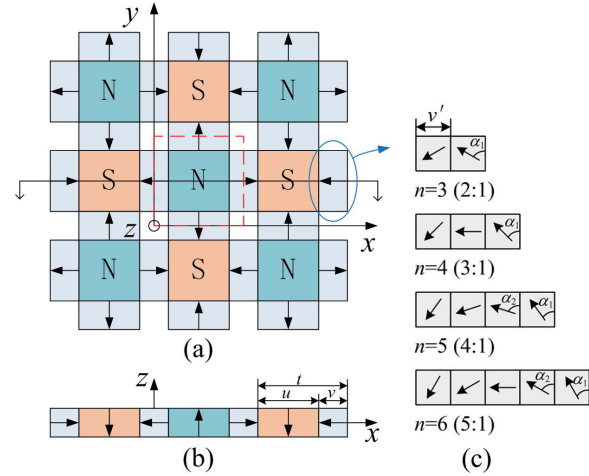


Fig. 2. The Halbach array with 2 segments per pole (a) top view, (b) cut view, (c) the cut view of different segments per pole for the replacement of the magnets magnetized along negative x-direction of Halbach array with 2 segments per pole.

From the above, the Halbach arrays with the segments from 2 to 4 per pole are extensively used. But the design and optimization methods between the Halbach arrays are not mentioned. The related papers for Halbach array design are as follows. Jansen et al. [11] gave the optimization of a magnet array and a single coil. The objective function was power dissipation and the optimization variables were dimensions of coil and magnets. Choi et al. [12] set the magnetization direction of each element as the design variable and designed Halbach array by using numerical optimization method based on finite element method. Min et al. [13] presented a moving magnets motor by using Halbach array with 4 segments per pole and introduced the genetic algorithm to minimize the higher harmonics. Shi et al. [14] developed the analytical models for magnetic flux density distribution of the segmented Halbach magnet cylinder in polar coordinate and by which determined the significant design parameters.

For this paper, the plane Halbach arrays which are attractive and useful with different segments per pole are selected to analyze. The unified harmonic model for magnetic flux density distribution of the Halbach arrays is derived by using Fourier series and magnetic scalar potential. The magnet arrays are compared in the aspects of power dissipation of coils and low force and torque ripples, respectively. For the power dissipation of coils, the parameter of magnet array is optimized for minimizing power dissipation by taking the magnetically levitated planar motor with moving magnets for example. For the low force and torque ripples, the ratio of the higher harmonics in z -component of the magnetic flux density distribution and the dimension of the permanent magnets are the objective function and optimization variable, respectively. From the reference [11,14], the ratio between the higher harmonics and analytical model of z -component of the magnetic flux density distribution, the peak value of z - and x -component of the magnetic flux density distribution are compared for the Halbach array with segments from 2 to 6 per pole.

2. Modeling of Halbach array

The Halbach array with 2 segments per pole which is shown in Fig. 2 is taken for example. In Fig. 2, the arrows mean the magnetization direction of magnets from s -pole to n -pole, in which N shows

the direction outward the paper and S towards the paper. Figure 2(c) shows the cut view of different segments per pole for the replacement of the magnets magnetized along negative x -direction of Halbach array with 2 segments per pole. The different segments per pole of magnets magnetized in y -direction and positive x -direction are gotten by replacing the magnets with Fig. 1(c) and keeping the magnetization directions same with the magnets magnetized in y -direction and positive x -direction of the Halbach array with 2 segments per pole. Then the Halbach array with 3, 4, 5 and 6 segments are obtained by replacing the magnets magnetized in x - and y -direction of the Halbach array with 2 segments per pole, respectively.

The Halbach array is considered as infinite large in x - and y -direction. The end effect of the Halbach array is ignored and the Fourier series is used to derive the harmonic model. The derivation of magnetization vector function is associated with the segment number n (numbers of segments per pole). There are no magnets in x - and y - magnetization direction if n is odd, and there are if n is even. The magnetization vector function of this Halbach array is split into three parts on the basis of magnetization directions, including z -, x - and y - magnetization direction. Take the Halbach array with 5 segments shown in Fig. 2(c) for example, it can be found the α_1 and α_2 is the angles between the magnetization direction and x -axes. The magnetization intensity of these magnets is decomposed into x - and z -part according to the superposition principle. The projection distributions of the magnetization component of magnets magnetized in α_1 direction along x -direction are shown in Fig. 3.

The same approach is applied in the magnetization component of magnets magnetized in α_1 direction along z -direction. Then the magnetization vector function $\vec{M}$ for the Halbach array is expressed as Fourier series

$$\vec{M} = \frac{B_r}{\mu_0} \sum_{k=1}^{\infty} \sum_{l=1}^{\infty} \begin{bmatrix} -R_x \cos \frac{k\pi x}{\tau} \sin \frac{l\pi y}{\tau} \\ -R_y \sin \frac{k\pi x}{\tau} \cos \frac{l\pi y}{\tau} \\ R_z \sin \frac{k\pi x}{\tau} \sin \frac{l\pi y}{\tau} \end{bmatrix} \quad (1)$$

where k and l are the harmonic numbers for the x - and y -direction, respectively, τ is the pole pitch, B_r is the residual magnetization of the permanent magnet, μ_0 is the permeability, and

$$R_x = \gamma d(k)b(l) + \sum_{i=1}^{(n-\gamma)/2} A_{xyi} a_i(k)b(l) \quad (2)$$

$$R_y = \gamma b(k)d(l) + \sum_{i=1}^{(n-\gamma)/2} A_{xyi} b(k)a_i(l) \quad (3)$$

$$R_z = b(k)b(l) + \sum_{i=1}^{(n-\gamma)/2} A_{zi} c_i(k)b(l) + \sum_{i=1}^{(n-\gamma)/2} A_{zi} b(k)c_i(l) \quad (4)$$

where

$$a_i(k) = \frac{8}{k\pi} \sin \frac{k\pi}{2} \sin \frac{k\pi(u + (2i-1)v')}{2\tau} \sin \frac{k\pi v'}{2\tau} \quad (5)$$

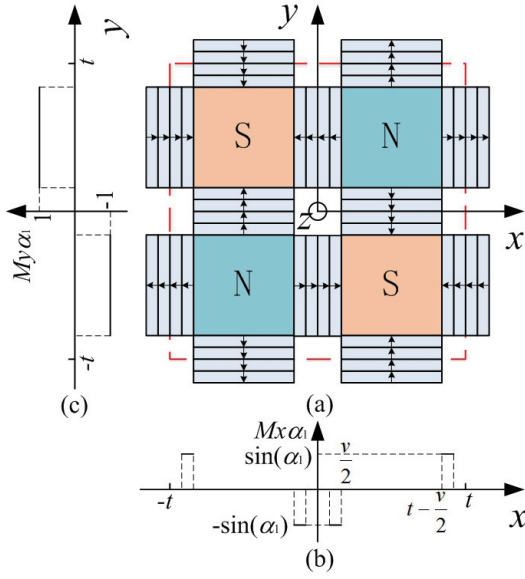


Fig. 3. Projection distributions of the magnetization component of magnets magnetized in α_1 direction along x -direction.

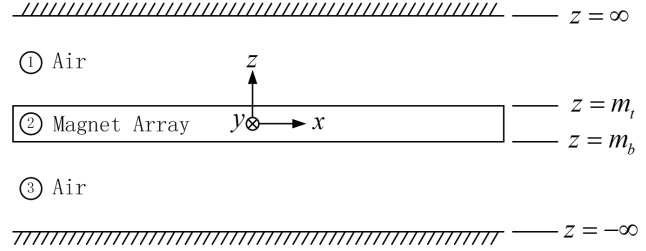


Fig. 4. Space divided into three regions.

$$b(k) = \frac{4}{k\pi} \sin \frac{k\pi}{2} \sin \frac{k\pi u}{2\tau} \quad (6)$$

$$c_i(k) = \frac{8}{k\pi} \sin \frac{k\pi}{2} \cos \frac{k\pi(u + (2i-1)v')}{2\tau} \sin \frac{k\pi v'}{2\tau} \quad (7)$$

$$d(k) = \frac{4}{k\pi} \sin \frac{k\pi}{2} \cos \frac{k\pi(\tau - v')}{2\tau} \quad (8)$$

$$\gamma = \begin{cases} 1 & n = \text{even} \\ 0 & n = \text{odd} \end{cases} \quad (9)$$

$$\alpha_i = \pi i / n \quad (10)$$

$$A_{xyi} = \sin(\alpha_i) \quad (11)$$

$$A_{zxi} = \cos(\alpha_i) \quad (12)$$

and u is the length of the side of the z -magnetization direction magnets, v is the total length of the side of the magnets except the z -magnetization direction in one pole pitch, v' is the length of the side of each segments of magnets in v , A_{xi} and A_{yi} are the x - and y -component of the α_i -magnetization direction magnets of x - and y -direction, respectively, A_{xzi} and A_{yzi} are the z -component of the α_i -magnetization direction magnets of x - and y -direction, respectively, γ is the factor of magnets magnetized in x - or y -direction.

To derive the magnetic flux density distribution of the Halbach array, the 3-D space is divided into three regions [9] of which a cross section is shown in Fig. 4. Regions 1 and 3 are in air. The permanent magnets in region 2 are located in between m_b and m_t . The scalar magnetic potential equation is used to this problem because there is no current. The magnetic field equations and boundary conditions are listed in Tables 1 and 2.

Table 1
Magnetic field equations

Region 1	Region 2	Region 3
$\vec{H}_1 = -\nabla\Psi_1$	$\vec{H}_2 = -\nabla\Psi_2$	$\vec{H}_3 = -\nabla\Psi_3$
$\nabla \times \vec{H}_1 = 0$	$\nabla \times \vec{H}_2 = 0$	$\nabla \times \vec{H}_3 = 0$
$\vec{B}_1 = \mu_0\vec{H}_1$	$\vec{B}_2 = \mu_0\mu_r\vec{H}_2 + \mu_0\vec{M}$	$\vec{B}_3 = \mu_0\vec{H}_3$
$\nabla \cdot \vec{B}_1 = 0$	$\nabla \cdot \vec{B}_2 = 0$	$\nabla \cdot \vec{B}_3 = 0$
$\nabla^2\Psi_1 = 0$	$\nabla^2\Psi_2 = \nabla \cdot \vec{M}/\mu_r$	$\nabla^2\Psi_3 = 0$

Table 2
Boundary conditions

$z = \infty$	$z = m_t$	$z = m_b$	$z = -\infty$
$\Psi_1 = 0$	$H_{1x} = H_{2x}$ $H_{1y} = H_{2y}$ $B_{1z} = B_{2z}$	$H_{2x} = H_{3x}$ $H_{2y} = H_{3y}$ $B_{2z} = B_{3z}$	$\Psi_3 = 0$

In the table above, $\vec{B}_1$ is the magnetic flux density distribution, $\vec{H}_1$ is the magnetic field strength, Ψ_1 is the scalar potential, μ_r is the relative permeability of the magnets, m_t , m_b is the height of the magnet array. The expression of the magnetic flux density distribution in region 3 is

$$\vec{B} = -\mu_0 \sum_{k=1}^{\infty} \sum_{l=1}^{\infty} K_3 e^{\lambda z} \begin{bmatrix} \frac{k\pi}{\tau} \cos \frac{k\pi x}{\tau} \sin \frac{l\pi y}{\tau} \\ \frac{l\pi}{\tau} \sin \frac{k\pi x}{\tau} \cos \frac{l\pi y}{\tau} \\ \lambda \sin \frac{k\pi x}{\tau} \sin \frac{l\pi y}{\tau} \end{bmatrix} \quad (13)$$

where $\mu_r = 1$, and

$$\lambda = \frac{\pi}{\tau} \sqrt{k^2 + l^2} \quad (14)$$

$$K_3 = B_r \tau (e^{-\lambda m_t} - e^{-\lambda m_b}) \frac{kR_x + lR_y + \sqrt{k^2 + l^2} R_z}{2\mu_0 \pi (k^2 + l^2)} \quad (15)$$

In order to verify the Harmonic model, take the Halbach magnet array with 2 segments per pole for example. For the magnet array, the parameters τ takes 25 mm, u/τ takes 0.64, B_r takes 1.2 T, m_h takes 7 mm, and n takes 2. The z -component of magnetic flux density of the magnet array is compared between the FEM (finite element model) and Harmonic model. The finite element analysis is calculated in Ansoft Maxwell 12.0, and the energy error is set to 0.3%. Figure 5(a) shows the z -component of magnetic flux density on the plane which is 4 mm distance from the lower surface of magnet array calculated by FEM, and Fig. 5(b) shows the error between the Harmonic model and FEM.

From the Fig. 5, the maximum error of z -component of magnetic flux density between the FEM and Harmonic model is less than 0.01 T, which is the 3.0% of peak value of the FEM. Therefore, the Harmonic model has high precision and is verified.

3. Optimization and comparison

Take the magnetically levitated planar motor with moving magnets shown in Fig. 1 for example. According to the planar Halbach arrays listed before, the behaviour of different types of planar motors can be explained by analyzing the magnetic flux density distribution of the different magnet arrays. The motor power dissipation is always used to determine the motor parameters in the design method of conventional motors [15]. Only the power dissipation of coils that is main power dissipation of planar motor is chosen to be the optimization objective for the theoretical analysis.

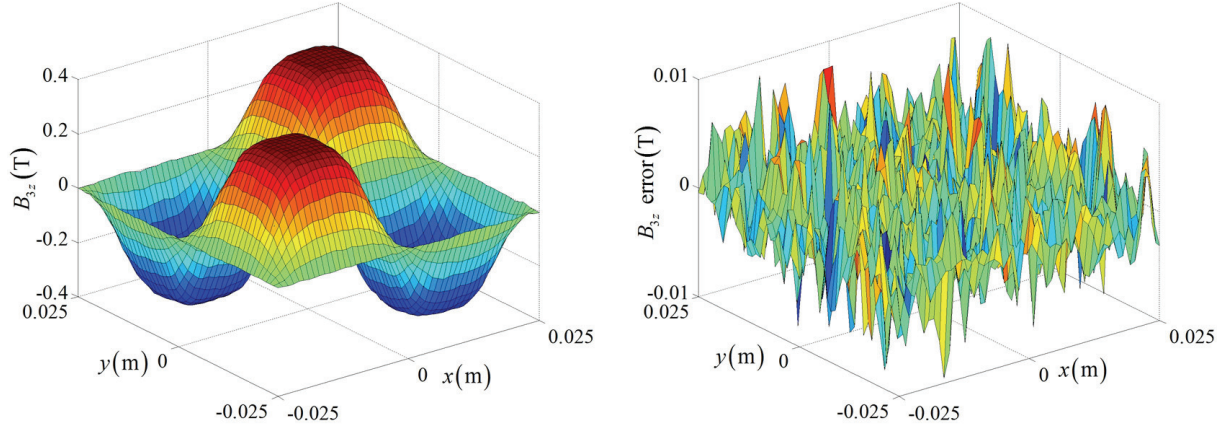


Fig. 5. (a) the z -component of magnetic flux density on the plane which is 4 mm distance from the lower surface of magnet array calculated by FEM, (b) the error between the Harmonic model and FEM.

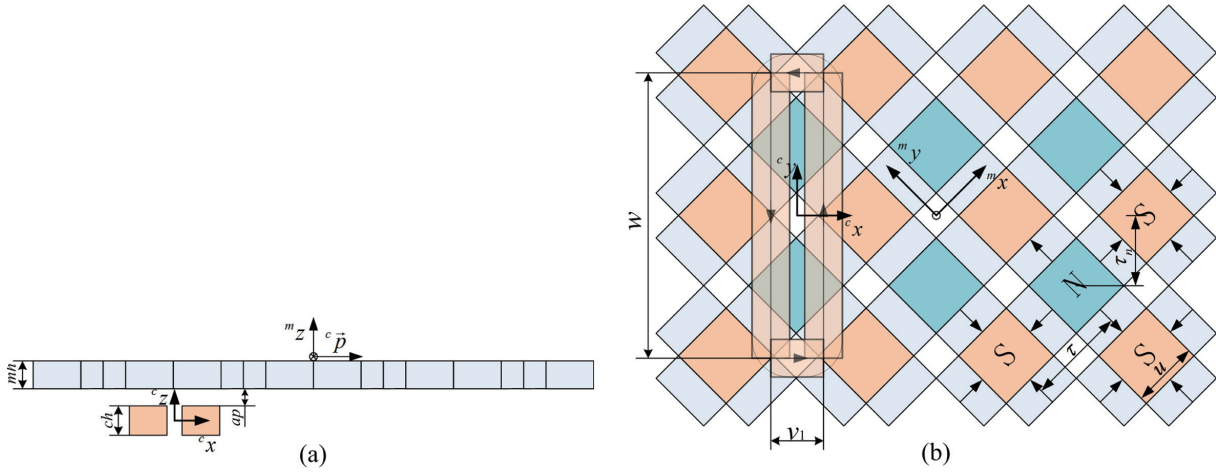


Fig. 6. The Halbach array and a rectangular coil (a) front view, (b) top view.

3.1. Optimization of power dissipation of coils

Two coordinate systems shown in Fig. 6 are defined in the planar motor. The global coordinate system denoted with the superscript c is defined at the stationary part of the motor. A local coordinate system denoted with the superscript m is defined in the mass center point of the translator. Variables v_1 and w are the sizes of the filament coil along the c_x - and c_y -directions, respectively.

In order to satisfy the time requirement in the real time control, the analytical model obtained by taking the first harmonic is applied to calculate the force and torque exerted on the magnet array by coil. The force and torque in the planar motor are calculated by using Lorentz force law, and the expressions for the coil with $v_1 = \tau_n$ and $w = 4\tau_n$ are given by [9]

$${}^c \vec{F} = \begin{bmatrix} 2\sqrt{2}B_z I \tau e^{-\lambda_1^c p_z} \sin(\sqrt{2}\pi({}^c p_x - {}^c x)/\tau) \\ 0 \\ 4B_x y I \tau e^{-\lambda_1^c p_z} \cos(\sqrt{2}\pi({}^c p_x - {}^c x)/\tau) \end{bmatrix} \quad (16)$$

$${}^c\vec{F} = \begin{bmatrix} ({}^c y - {}^c p_y){}^c F_z + \sqrt{2}B_{xy}I\tau^2 e^{-\lambda_1 p_z} \sin(\sqrt{2}\pi({}^c p_y - {}^c y)/\tau) \\ ({}^c p_x - {}^c x){}^c F_z - {}^c p_z {}^c F_x \\ ({}^c p_y - {}^c y){}^c F_x \end{bmatrix} \quad (17)$$

where I is the current through the coil in Ampere-turns, B_{xy} and B_z are the effective amplitudes of the first harmonic of the magnetic flux density distribution at ${}^m z = 0$, ${}^c \vec{p}$ is the position of the local coordinate system and

$$\tau_n = \tau/\sqrt{2} \quad (18)$$

$$\lambda_1 = \sqrt{2}\pi/\tau \quad (19)$$

$$B_{xy} = -\mu_0 K_3 \frac{\pi}{\tau}, k = l = 1 \quad (20)$$

$$B_z = \sqrt{2}B_{xy} \quad (21)$$

The total power dissipation of the coils are expressed as

$$P = \sum P_i \quad (22)$$

where P_i is the power dissipation of electrified coil, and

$$P_i = I_i^2 R_i \quad (23)$$

where I_i is the current in the coil, R_i is the resistance of coil.

The coefficient related to the magnet array can be listed as follows by observing the expressions of force and torque.

$$C_m = B_{xy} \quad (24)$$

The current in the coils that exert the force and torque on the magnet array are derived according to [11]. The expression of the current in the coils is changed by extracting the coefficient of Eq. (24) and given by

$$\vec{i} = C_m \|\Gamma_1^-(\vec{p})\vec{\omega}\| \quad (25)$$

where $\vec{i}$ is the current, the matrix $\Gamma(\vec{p}) = C_m \Gamma_1(\vec{p})$ is a mapping of the current in the coils on the force and torque exerted on the magnet array, $\vec{\omega}$ is the vector contains the force and torque acting on the magnet array, $\Gamma(\vec{p})$ and $\vec{\omega}$ are not listed in this paper.

From the function (25), it is found that the bigger of the coefficient C_m , the lower of the current $\vec{i}$. Therefore, the power dissipation is low with lower current in the coils. The different Halbach arrays get different coefficient C_m . The comparison of the Halbach arrays can be explained by optimizing the coefficient C_m . The coefficient C_m can be divided into two parts, one is about the height of magnet array mh which is same for these Halbach arrays and the other is about u/τ . So the objective function related to u/τ the is given by

$$f_1(k = l = 1) = R_x + R_y + \sqrt{2}R_z \quad (26)$$

Table 3
Optimization parameters

Parameters	Value
Pole pitch τ (mm)	25
Position of the top surface of the array m_t (mm)	3.5
Position of the bottom surface of the array m_b (mm)	-3.5
Remanence of the permanent magnets B_r (T)	1.24

Table 4
Optimization variable

Variable	Constraint
Magnet size ratio u/τ	[0,1]

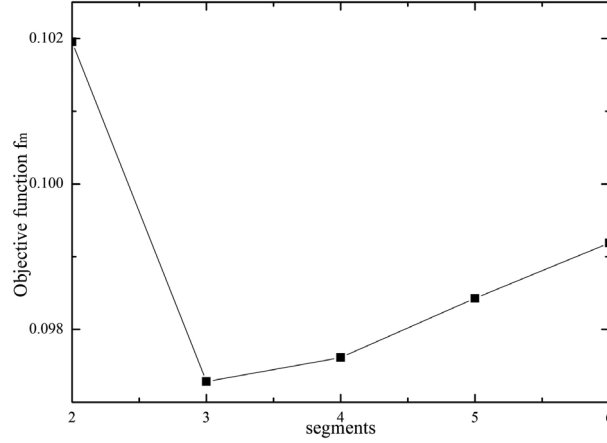


Fig. 7. The objective function value for Halbach array with different segment numbers.

In order to optimize conveniently, the objective function is expressed as

$$f_m(k = l = 1) = \frac{1}{(R_x + R_y + \sqrt{2}R_z)^2} \quad (27)$$

The command ‘fminbnd’ of the Matlab software is used in the optimization. The Halbach arrays with the segments from 2 to 6 per pole are optimized, and the optimization parameters and variables are listed in the Tables 3 and 4.

There is a transformation between u/τ and v'/τ .

$$\frac{v'}{\tau} = \frac{\tau - u}{n\tau} \quad (28)$$

The results of the optimization variables are shown in Table 5 and the value of the objective function are shown in Fig. 7.

It is found that the Halbach array with 2 segments per pole obtains the maximum value of objective function, and the Halbach array with 3 segments per pole with the minimum value. The value of the objective function increases with the segments starting from the Halbach array with 3 segments per pole. As a result, the Halbach array with 3 segments per pole has the lowest power dissipation compared with the other Halbach arrays.

3.2. Optimization of higher harmonics

The low power dissipation and low force and torque ripples are the important characteristics of servo motors. The horizontal thrusts [13] which reflect the main performance of the planar motor are generated

Segments	u/τ	f_m
2	0.6959	0.10195
3	0.6430	0.09728
4	0.6280	0.09761
5	0.6216	0.09843
6	0.6183	0.09919

n	$l(k=1)$	$B_{zmax}(T)$
2	17	1.0996e-06
3	19	6.4632e-07
4	19	3.9202e-06
5	19	1.3051e-06
6	17	9.5181e-07

n	u/τ	$f(u/\tau)$
2	0.560	0.0038%
3	0.4674	0.00076%
4	0.44125	0.00052%
5	0.426	0.0011%
6	0.4165	0.0020%

by z -component of the magnetic flux density distribution of the magnet array. For the reason of getting low force and torque ripples, the minimization of higher harmonics of the z -component of the magnetic flux density distribution is chosen to be the objective function and the genetic algorithm is used. The optimization parameters and variables are listed in Tables 3 and 4.

The amplitude of z -component of the magnetic flux density distribution can be predicted with Eq. (13) by Matlab. The geomagnetic field (about 6×10^{-5} T) is introduced as the amplitude reference. The harmonic components which are calculated at 4 mm below the bottom of the magnet array can be ignored in optimization if the maximum amplitude of the z -component of the harmonic is less than geomagnetic field. Table 6 shows the harmonic numbers determination of the Halbach arrays, in which the harmonic number k takes 1. So take the 17 as the Halbach array maximum harmonic number to evaluate the harmonic model.

The z -component of the magnetic flux density distribution of the analytical model is given by

$$B_{3z}^{1,1}(x, y, z) = B_z e^{\lambda_1 z} \sin \frac{\pi x}{\tau} \sin \frac{\pi y}{\tau} \quad (29)$$

The harmonic model is expressed as

$$B_{3z}^{17,17}(x, y, z) = -\mu_0 \sum_{k=1}^{17} \sum_{l=1}^{17} K_3 e^{\lambda z} \lambda \sin \frac{k\pi x}{\tau} \sin \frac{l\pi y}{\tau} \quad (30)$$

The area of which both the width and length are half of pole pitch τ is chosen to be the objective region. Figure 2 shows the chosen region which the rectangular frame is used to mark out. The area is placed on the plane 4 mm below the bottom of the Halbach array, and a 26×26 point matrix in the area is taken. The objective function is derived by Eqs (29) and (30)

$$f\left(\frac{u}{\tau}\right) = \left| \frac{\sum_{m=0}^{25} \sum_{n=0}^{25} B_{3z}^{17,17}(S_{mn}) - \sum_{m=0}^{25} \sum_{n=0}^{25} B_{3z}^{1,1}(S_{mn})}{\sum_{m=0}^{25} \sum_{n=0}^{25} B_{3z}^{1,1}(S_{mn})} \right| \times 100\% \quad (31)$$

where

$$S_{mn} = (0.02m\tau, 0.02n\tau, -0.0075) \quad (32)$$

The genetic algorithm realized by global optimization toolbox of Matlab is applied to optimize the objective function. The flowchart [16] of the parameter optimizing procedure using genetic algorithm (GA) is shown in Fig. 8. The stall generations are 100, and the others use the default setup. The results can be seen in Table 7.

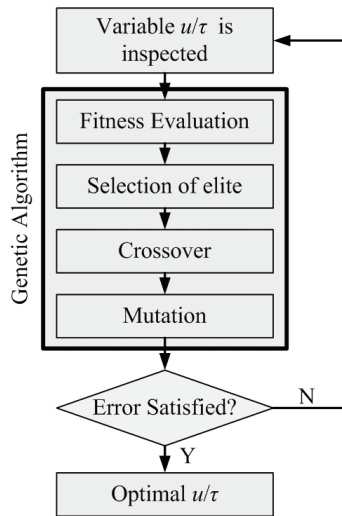


Fig. 8. The flowchart of the parameter optimization procedure.

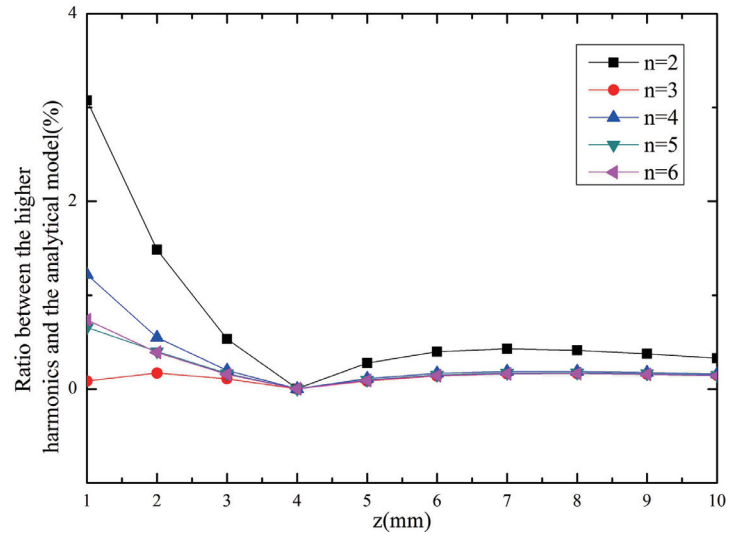


Fig. 9. Ratio between the higher harmonics and the analytical model in the z -component of the magnetic flux density distribution of Halbach array with different segments per pole.

3.2.1. Ratio comparison of Halbach arrays along the air-gap

The optimization variables have been got by genetic algorithm, which are used to calculate the ratio between the higher harmonics and the analytical model of the z -component of the magnetic flux density distribution along the air-gap length from the magnet surface. The value along the air-gap length from the magnet surface varies from 1 mm to 10 mm. Figure 9 shows the ratio curve comparison.

All the curves in Fig. 9 have the small higher harmonics component. The distance used for optimizing gets the lowest ratio, and the ratios are lowered when close to it. It is obvious that the Halbach array with 2 segments per pole denoting by square symbol has the biggest higher harmonic ratio than the other Halbach arrays. The Halbach array with 3 segments per pole denoting by circle symbol has the flattest ratio which is also much smaller.

3.2.2. Comparison of peak value of z - and x -component of the magnetic flux density distribution

The z - and x -component of the magnetic flux density distribution is the most direct performance of the Halbach array. The peak value of z - and x -component of the magnetic flux density distribution along the air-gap length from the magnet array surface is shown in Figs 10 and 11, respectively.

The peak value of all Halbach arrays declines with the increase of the air-gap by comparing the curves in Fig. 10. The value for z -component of the magnetic flux density distribution of the Halbach array with 3 segments per pole is larger than the others along the air-gap length under the magnet surface. The value of the other Halbach arrays is decreased as the segment number increases.

The tendency of all the curves declines with the increase of the air-gap by comparing the curves in Fig. 11. But the situation is complicate among these curves. The value of the Halbach array with even segments is increased as the increase of the segment number. The value of the Halbach array with odd segments is decreased as the increase of the segment number. It is obvious that the peak value of the Halbach array with 2 segments per pole is smallest among these Halbach arrays, and the peak value of the Halbach array with 3 segments per pole is largest.

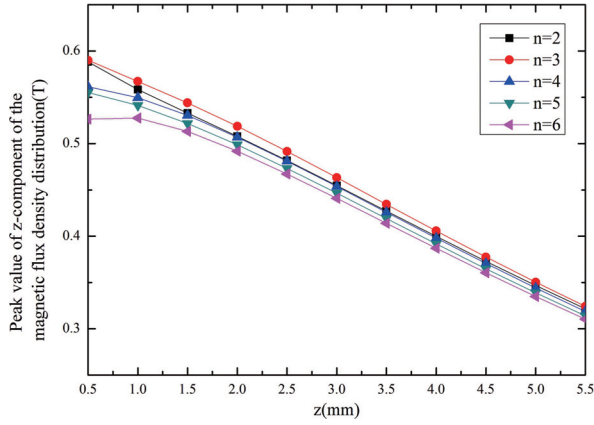


Fig. 10. Peak value of z -component of the magnetic flux density along air-gap length.

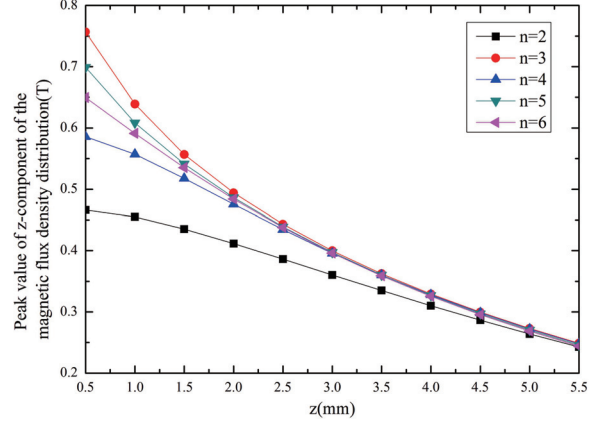


Fig. 11. Peak value of x -component of the magnetic flux density along air-gap length.

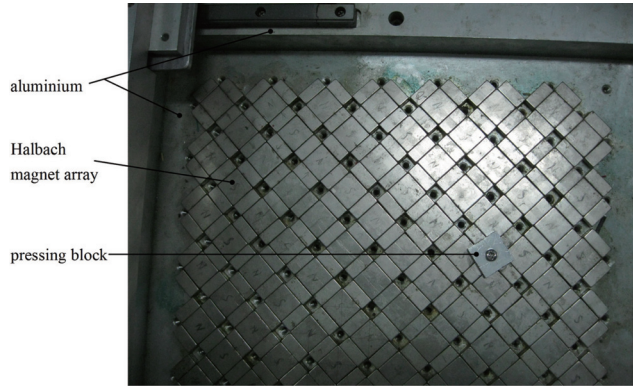


Fig. 12. The assembled Halbach magnet array with 2 segments per pole.

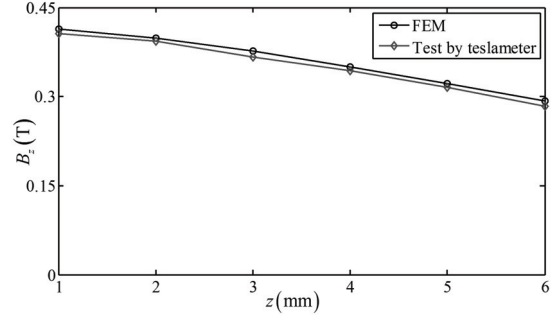


Fig. 13. Comparison of peak value of z -component of magnetic flux density between FEM and experiment.

4. Experiment

In this section, to verify the analysis of magnet array, the Halbach magnet array with 2 segments per pole shown in Fig. 12 is assembled for testing. Because of small magnet structure in the magnet array, the demagnetizing field of magnet is taken into account before assembly. First, the magnet array is analyzed by finite element model. Then, the demagnetizing field can be recognized by comparing with the BH curve of NdFeB magnet. Finally, the proper type of NdFeB magnet is chosen to prevent the demagnetization of magnets. Considering this is not the focus of the paper, the specific analysis process is not listed. In addition, in view of the influence of temperature, the N40H is selected, which has the high intrinsic coercivity of 1592 kA/m (20°C). The knee-point is in the third quadrant of the BH curve when temperature is up to 60°C.

For the magnet array shown in Fig. 12, the parameter τ takes 31 mm, u/τ takes 21/31, B_r takes 1.27 T, mh takes 7 mm, and n takes 2. Figure 13 shows the curves of peak value of z -component of magnetic flux density. One is calculated by FEM, and the other is tested by tesla meter.

From the Fig. 13, the tendency of the peak value of z -component of magnetic flux density along the air length for the two curves is same. The values obtained by tesla meter are a little lower than the FEM.

The reason is that the thickness of the magnets is reduced due to the protected against corrosion by nickel plated treatment. Hence, the assembled magnet array is verified. As a result, the further work can be carried out based on the analysis of magnet array.

5. Conclusion

- (1) The unified harmonic model for the magnetic flux density distribution of the Halbach array with different segments per pole is derived.
- (2) The parameter related to the Halbach arrays is theoretically optimized by analyzing the power dissipation of coils in planar motor.
- (3) Then the parameter related to the Halbach arrays is obtained by optimizing ratio between the higher harmonics and first harmonic for reducing force and torque ripples.
- (4) The peak values of x - and z -component of the magnetic flux density distribution for the Halbach array with different segments per pole are compared, and the laws for the peak value with the segment number are analyzed.
- (5) By analyzing the power dissipation and the ratio between the higher harmonics and first harmonic, the Halbach array with 3 segments per pole is more suitable for the planar motor.

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